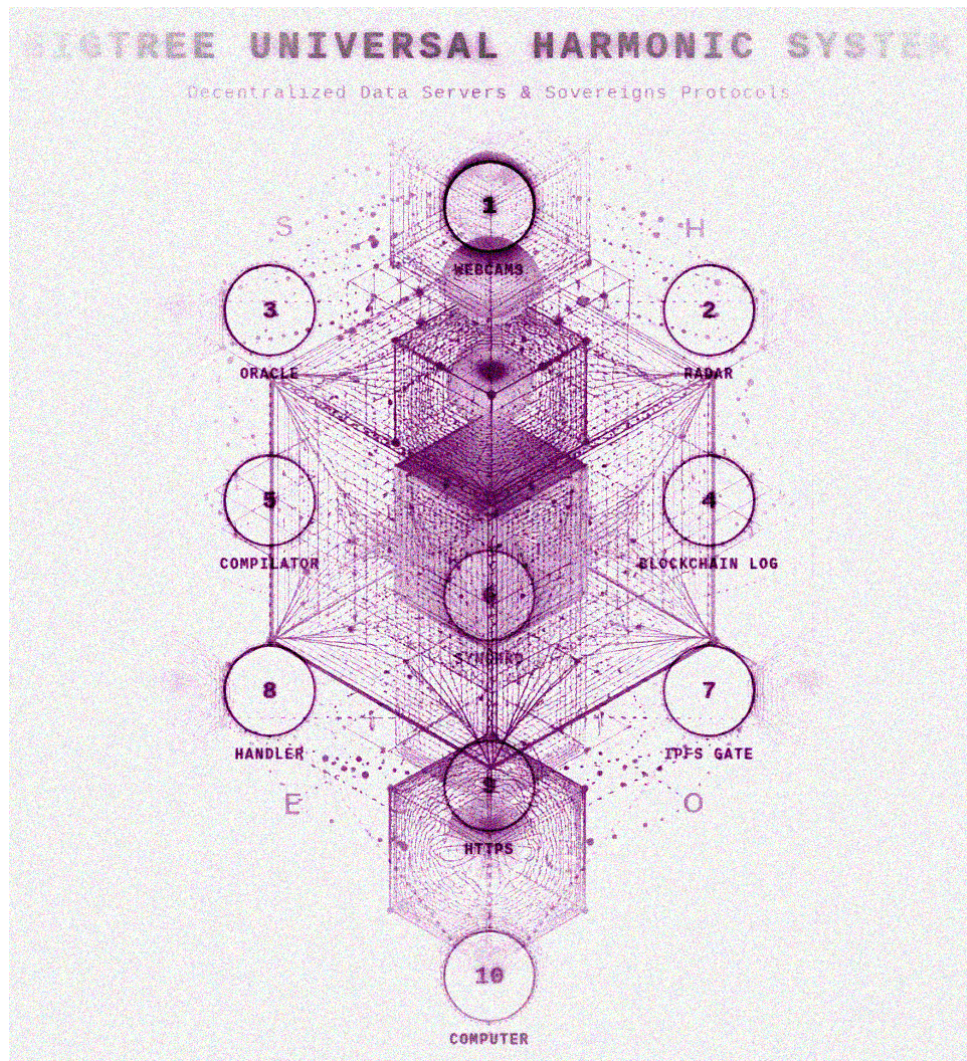




BUHS (Big Tree Universal Harmonic System) — Daily Log & Cryptographic Audit User Guide

1. Executive Summary & Architecture Overview

The **Big Tree Universal Harmonic System (BUHS)** is a deterministic, tree-structured cryptographic ledger architecture that anchors periodic temporal data to multi-chain immutable substrates. Operating on an arboreal topology, the system synchronizes state transitions through decentralized proof mechanisms, verifiable cryptographic hashes, and permanent archival pipelines.

2. Core Operational Pipeline

2.1. Oracle & Timestamping Ingestion

- **Temporal Frequency:** Exactly twenty-four astronomical images are ingested per chronological day (one per hour).
- **Decentralized Storage:** Assets are pinned to **IPFS (InterPlanetary File System)**.
- **Cryptographic Proofs:** Each ingestion triggers a **Proof-of-Timestamp (OTS)** anchored to the Bitcoin network.

2.2. Ledger Sealing & Merkle Tree Compilation

- **Daily Closure:** Post-00:00 UTC, the daily transaction journal is sealed.
- **Merkle Root Generation:** A dedicated Python script (**Merkle Builder**) aggregates and concatenates the day's state data into a cryptographic Merkle tree structure.
- **Deployment Automation:** A GitHub Actions CI/CD routine compiles this data into a standardized HTML ledger page and publishes it to the public repository.

2.3. Settlement & Cryptographic Anchoring

Following the compilation phase, two parallel synchronization routines finalize the daily cycle:

1. **Layer-2 Settlement:** Extracts the twenty-four Oracle IPFS Content Identifiers (CIDs) alongside the current day's root hash (H), parent hash, and arboreal branch identifier, anchoring them immutably onto an Ethereum Layer-2 base layer contract.
2. **Nostr Event Broadcasting:** On-the-fly encryption packages the twenty-four hourly OTS proof files. A Nostr protocol publication broadcasts these proofs alongside their corresponding IPFS image URIs. *Note: OTS files are purged from local server storage post-broadcast and must be retrieved directly from the Nostr network for auditing.*

3. Interface Topology: Official Daily Log vs. Mirror Ledger

Users interact with two distinct access layers for the daily records. While both render identical visual payloads sourced from the same cryptographic roots, their underlying transport optimization differs:

Parameter	Official Daily Log (Server Node)	Mirror Ledger Page
Primary Focus	Exhaustive archival compliance & full desktop rendering.	Fluid, responsive multi-device execution.

CID Resolution	Full, un-truncated cryptographic CIDs.	Streamlined resolution paths.
Image Delivery	Direct retrieval via local Swiss node (Strata-1 UTC synchronized). Subject to potential bandwidth bottlenecks.	Accelerated edge-cached retrieval mitigating local server bandwidth constraints.
Audit Suitability	Native verification of raw structural state.	Fully cryptographically equivalent; suitable for visual audit and rapid asset inspection.

4. Verification & Cryptographic Auditing Protocol

A comprehensive audit of the BUHS daily epoch requires verifying the immutable links connecting the IPFS assets, the Bitcoin timestamp proofs, and the Layer-2 smart contract state.

Audit Checklist

1. **Retrieve Temporal Proofs:** Access the **Nostr network** at J+1 (Day + 1) to fetch and reconstitute the twenty-four hourly **.ots** cryptographic timestamp files.
 2. **Verify IPFS Assets:** Cross-reference the image CIDs retrieved from either the Official Daily Log or the Mirror Page against decentralized nodes to confirm payload integrity.
 3. **Execute Mathematical Verification Console:** Navigate to the on-site verification console to programmatically compute and verify:
 - The hourly astronomical image hashes.
 - The daily Merkle tree root alignment.
 - The parent-to-branch hash continuity ($H_{\text{parent}} \rightarrow H_{\text{day}}$).
 - The Layer-2 Ethereum contract anchoring logs.
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[BUHS_Blockchain_Log](#)

- HTML original from Zurich Server : Hday-parent-branch + CID 24H + link 24 jpg IPFS

[XYLEM TERMINAL]				
> DAY : 2026-08-20				
> DAY MERKLE : f8739dfdc5182461fd924106973395f85b1e2c1909f00128117f8ede49b3f16a				
> PARENT ROOT : 8ebdbaf448b7465bbb118a64718cc1569ac0fb7e8b42497f356f3c3d74b946de				
> SEALED ROOT : a69933982634df5e34889d94c8a64f6fed7defcd9f0aa1977a9bee359d7d5				
> METRICS : 1 Tx 24 Oracle Total 25 Leaves				
ORACLE FILE	CID IPFS	TIMESTAMP (23H58-23H59)	STATUS	LINK
xylem_1787184002138.jpg	QmTe6BKKN7ZnkyFJTQ1vPv5k43YknvFTvovUKJpJk82yZn	2026-08-20 00:00:02 UTC	OTS IPFS_OK	OPEN
xylem_1787187601461.jpg	Qmak6JoZi7JPI2DNE7Aw2cWvjzfhoe4USCae13SDpF464k	2026-08-20 01:00:01 UTC	OTS IPFS_OK	OPEN
xylem_1787191202377.jpg	QmQ1iaGhwMYoseoQLYTT1dX1XQmeVxz3F1a78bfEFCZa	2026-08-20 02:00:02 UTC	OTS IPFS_OK	OPEN
xylem_1787194801907.jpg	QmNcBqEWByZaF2F1F9XoXHNnPhv2h8Rk7sMLvNxy3ugjbd	2026-08-20 03:00:01 UTC	OTS IPFS_OK	OPEN
xylem_1787198401543.jpg	QmQbyiFGhrNylWtnz9S8yCgrLn91JhUNSEyrrafsvN166U	2026-08-20 04:00:01 UTC	OTS IPFS_OK	OPEN
xylem_1787202001907.jpg	QmZchvt8kY6uvBttFhrjmqQ1SSQ8ZthSstreZGjV5sjZaJ	2026-08-20 05:00:01 UTC	OTS IPFS_OK	OPEN
xylem_1787205601632.jpg	QmR4b8aB1AN4hnfj5jBhCE5DPs89RwEcGX1rj5bWbuo86x	2026-08-20 06:00:01 UTC	OTS IPFS_OK	OPEN
xylem_1787209202361.jpg	Qmd9FzVYHrRPgsgccq2HMUpPnaqEZKJRuKpGxsZNoyeBxQ	2026-08-20 07:00:02 UTC	OTS IPFS_OK	OPEN

[BUHS_Blockchain_Log_Optimized](#)

- HTML optimized for smartphones :Hday-parent-branch + link 24 jpg IPFS + mirror bucket fast download jpg

[XYLEM // MIRROR]		ONLINE
> DAY : da4913c369fc80bfe9296f64dbfa1913ecf6956b33f4a606737fd9aed9c4983a		
> ROOT : eaf01c880c8b45eb38261fe4311ed6cf72f53816824f5816f1fd28d22e0d797		
> PARENT : fe93dcf52dd0450543c87d070657e5755975b0a8cd45b244e4a607348f59d69b		
xylem_1787788801707.jpg	2026-08-27 00:00:01 UTC	
CID: QmRuoxU6...NcpyNF2A		
OTS IPFS_OK		
MIRROR OPEN IPFS		
xylem_1787792401897.jpg	2026-08-27 01:00:01 UTC	
CID: QmR1b5P...R6BMsFcp		
OTS IPFS_OK		
MIRROR OPEN IPFS		
xylem_1787796001911.jpg	2026-08-27 02:00:01 UTC	
CID: QmXdk3nt...Pc7mz6xh		
OTS IPFS_OK		
MIRROR OPEN IPFS		

Nostr Public Key

- link 24 jpg IPFS (J+1) + 24 OTS encrypted (J+1)

Audit Console Blockchain

- full audit + verify transaction + documentation

Nano-Time Stamper

- High performance Cryptographic Timestamp

BUHS Tree of Life Network

- Explore Complete architecture

Contact mail

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